R.D. DUTTA & CO. PVT. LTD.

302, Rattan Jyoti Building 18,
Rajendra Place, New Delhi - 110008
Phone : 25710739, 25739805
Email : info@rddutta.com
Web : www.rddutta.com
CIN No. : U74899DL1982PTC014209

Quotation No : Q24000610

Date : 01-Jul-2025

Sidwal Refrigeration Industries Pvt Ltd
Plot No. 23, Sec-6
FARIDABAD
HARYANA

Kind Attention : Mr./Ms. Uma Shankar

Dear Sir,

It's been more than 30 years since R D Dutta & Company came into existence in the field of Compressed air systems and accessories.

We have an expertise in Providing best of sales and after sales support and a wide range of products to meet our customer needs.

To us being leader not only means selling the best quality products but, it also means working hard to understand your industry, your needs and the demands placed on your productivity in order to provide you with the most beneficial solutions for your work.
We trust you Find our offer in line with your requirement.

Should you require any further clarification/information, please contact the undersigned.
Thanking you of out best attention at all times.

Very truly yours,

For R. D. Dutta & Co. PVT. LTD.

Rakesh Dutta

Director

9311955930

DT1005



nGEN POWERED BY ABSSTEM INDUSTRIAL

Industrial

PSA Nitrogen Generator

Unique Features of Absstem Nitrogen Generator nGEN UH

a. **Lowest Air to Gas ratio** in industry thereby guaranteeing **lowest power consumption**.
b. **10 years*** warranty on Molecular Sieves **(First In Industry)**.
c. Assured Performance for consistent Flow, Pressure and Purity of Nitrogen throughout Product Lifecycle.
d. IIOT ready machine having 100% operator free design for 24X7 operation.
e. Idle mode with auto start/stop features ensuring power savings during low/no demand.
f. Inbuilt dew point transmitter to monitor input compressed air quality as standard scope of supply.
g. Plug n Play machine with an ergonomic design suitable for accessibility.
h. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
i. All critical performance parameters are being monitored through HMI.
j. First in industry **Comprehensive Maintenance Contract (CMC)** service under Absstem Premium Shield program providing complete peace of mind to customer.

Ambient Design Parameters

a.	Temperature	0 to 45 Deg. Celsius
b.	Relative Humidity	up to 80 % (Max.)
c.	Altitude	up to 1000 m above Sea level

Absstem Industrial Solutions Pvt. Ltd.
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

Product Description

a. Product Name	Ultra Purity PSA Nitrogen Gas Generator
b. PSA Module	**nGEN UH**
c. Purity Range	99.99% - 99.999%

General PSA Generation System Layout

A typical Industrial Nitrogen Generation system consists of three parts:

- Compressed Air System: Air Compressor, Pre-filter, Air Dryer, Dry Air Vessel
- PSA N_2 Generation System
- N_2 Storage Tank



Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air.
Refrigeration dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Tank	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air.
Storage Tank	Industrial Grade **Nitrogen** produced from the PSA System is stored in this for continuous supply.
Oxygen Analyser	Nitrogen produced is monitored by an Oxygen analyser. In case, O_2 impurity goes above the specified level, it gives an alarm and switches the plant to vent mode.

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HD Touch Screen Display

(User-friendly touch screen interface for easy operation and maintenance)

Absstem industrial Nitrogen generators complies with all industry standards and regulation

- Alarm management system
- Process data display
- Consumables running hours' display
- Data logging via USB drive**
- Graphical representation**



Additional upgrades

- Digital flow meter with totalizer
- Nitrogen Pressure monitoring for High pressure booster driven plants
- Cylinder manifold and alarm management system
- Remote monitoring
- Nitrogen gas Dew-point monitoring

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Gas generators.

TÜV NORD ISO 9001 · TÜV NORD ISO 14001 · TÜV NORD ISO 13485 · RIR CERTIFICATION Partners in Growth

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Absstem Shield Premium

(Comprehensive Annual Maintenance Contract)



Any Industry is now heavily focused on increased OEE (Overall Equipment Efficiency) and reduced LCC (Life Cycle Cost) while deciding to invest in any equipment. At Absstem, we strive for an everlasting engagement with our customers by not selling them just a product but also provide a long term after-sales support.

We understand that to deliver a consistent performance, periodic health check and timely maintenance of the Nitrogen Generator is extremely important. With Absstem Shield Premium, you invest in complete peace of mind for all maintenance needs pertaining to N2 generators and associated integrated equipment like air compressor, Air dryer etc.

Few benefits of Absstem Shield Premium are listed as below: -

- Guaranteed assurance of plant performance.
- Genuine spare parts and consumables being used to provide longevity and durability.
- Regular visits by our technicians for health monitoring and carrying out the scheduled maintenance services.
- 24 x 7 On-Call service support to our customers for any basic trouble shooting
- Emergency visit to attend any unplanned breakdown to any location within 48 hours.
- No additional surprise charges as all breakdown spares and consumables covered.
- The contract can be signed at the time of equipment ordering itself and can be renewed on year on year basis.

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Product Description

Product	Description	Make	Qty
Compressed Air System	C022E0145 Air System		1

Caption	Specification
Air Compressor KW	22 KW
Air Compressor Make	Atlas Copco/ Ingersoll Rand/ Gardner Denver/ As per Demand
Air Dryer	Included
Dry Air Storage	1500 Liters Included
Interconnecting Piping	Included
Pre filter	Included
Dry Air Generated	145 CFM
Air Pressure	7.5 Barg
POWER SUPPLY	400 V 3 Phase 50 Hz

Product	Description	Make	Qty
nGen Nitrogen Generator Ultra Purity	nGEN055UH Nitrogen		1

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	upto 6.5 Barg
FLOW AT 99.99% PURITY	52 Nm3/Hr DRY AIR REQUIRED 147 CFM
FLOW AT 99.995% PURITY	47 Nm3/Hr DRY AIR REQUIRED 165 CFM
FLOW AT 99.999% PURITY	35 Nm3/Hr DRY AIR REQUIRED 165 CFM
COMPRESSED AIR PRESSURE REQUIRED	7.5 Barg
RECOMMENDED NITROGEN STORAGE	1500 Liters
POWER SUPPLY	230 V/ 110 V 50/60 Hz Single Phase
RECOMMENDED AIR RECEIVER	1500 Litres Part of Air System
INPUT AIR DEW POINT MONITOR	INCLUDED
CMS Warranty	10 Years*

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Product	Description	Make	Qty
Gas Storage Vessel	AI-1500L Storage Vessel		1

Caption	Specification
Water Volume	1500 Litres
Gas Stored	Nitrogen/Oxygen
Design Standard	ASME Sec. VIII Div.1 Pressure Vessel Design
Working Pressure	8 Barg
Hydrotest Pressure	13.2 Barg
Accessories Included	Pressure Guage Safety Valve Manual Drain Additional Port for Other Instruments
Material of Construction	IS2062 Gr.B
Gas Stored Volume	9.75 M3
Corrosion Allowance	1.5 MM

Product	Description	Make	Qty
Lubricated Booster	IR-15T2X20 Booster	ABSSTEM	1

Caption	Specification
Gas to be compressed	Nitrogen
Suction Pressure	3 to 4 Barg
Discharge pressure	8 to 70 Barg
Motor rating	15 KW
Voltage	415 V AC
Gas Flow	51 to 62 NM3/hr
Type	Air Cooled Lubricated Reciprocating Booster

Product	Description	Make	Qty
HP Storage Tank 65 Barg	2000 ltr HP Storage Tank 65 Barg High Pressure Storage Tank		1

Caption	Specification
Volume	2000 litres
Working Pressure	65 Barg
Hydrotest Pressure	107.25 Barg

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Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	GST(%)	Amount
1	ACSC022E	C022E0145	Compressed Air System	840510	1	12,05,000.00	18.00	12,05,000.00
2	nGEN055UH	nGEN055UH	nGen Nitrogen Generator Ultra Purity	840510	1	36,39,000.00	18.00	36,39,000.00
3	AIS0158ST	AI-1500L	Gas Storage Vessel	731100	1	1,30,000.00	18.00	1,30,000.00
4	IR-15T2X20	IR-15T2X20	Lubricated Booster	8414	1	6,50,000.00	18.00	6,50,000.00
5	HPS16	2000 ltr HP Storage Tank 65 Barg	HP Storage Tank 65 Barg	73110090	1	8,92,500.00	18.00	8,92,500.00

Sub Total	65,16,500.00
GST(%) @18.00	11,72,970.00
Grand Total	76,89,470.00

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Terms And Conditions

1.	EXCLUSIONS & CLIENTS SCOPE	All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required).
2.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
3.	P&F	3% Extra.
4.	Freight	Extra at Actuals.
5.	Insurance	In your Scope.
6.	Payment Terms	50% Advance with your Prestegious Purchase Order and Balance against PI.
7.	Delivery	12-16 Weeks from the PO and Advance. Rest can be confirmed for Availability.
8.	Warranty	We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
9.	GST	As per the Commercial Table.
10.	Remarks	.

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For R. D. Dutta & Co. PVT. LTD.

Rakesh Dutta

Director

9311955930

DT1005

PSA Nitrogen Generator Comparison Report

Cost & Efficiency Analysis

Operational Parameters

Required Nitrogen Purity:	99.99%
Required Flow:	60 Nm³/hr
Power Cost:	₹8.2/kWh
Annual Operating Hours:	8000 hours

System Specifications

Absstem System

Available Nitrogen Flow:	52 Nm³/hr
Compressor Power:	26 kW
Utilization Factor:	0.87
Total Input Power:	22.50 kW
Specific Power:	0.43 kW/Nm³

Competition System

Available Nitrogen Flow:	60 Nm³/hr
Compressor Power:	45 kW
Utilization Factor:	1.00
Total Input Power:	45.00 kW
Specific Power:	0.75 kW/Nm³

Cost Analysis (10-Year Lifecycle)

Absstem System Costs

Equipment Cost:	₹65,00,000
Annual Power Cost:	₹17,03,298
10-Year Lifecycle Cost:	**₹2,35,32,982**

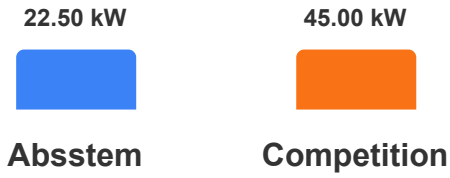
Competition System Costs

Equipment Cost:	₹40,00,000
Annual Power Cost:	₹29,52,000
10-Year Lifecycle Cost:	**₹3,35,20,000**

Savings Summary

Annual Power Cost Savings:	₹12,48,702
Total 10-Year Savings:	**₹99,87,018**
Return on Investment:	2.0 Years

Power Consumption Comparison



Power Savings: 22.50 kW

Annual Power Cost Comparison



Annual Savings: ₹12,48,702

10-Year Cost Projection

Year	Absstem	Competition	Savings
0	₹65,00,000	₹40,00,000	-₹25,00,000
1	₹82,03,298	₹69,52,000	-₹12,51,298
2	₹99,06,596	₹99,04,000	-₹2,596
3	₹1,16,09,895	₹1,28,56,000	₹12,46,105
4	₹1,33,13,193	₹1,58,08,000	₹24,94,807
5	₹1,50,16,491	₹1,87,60,000	₹37,43,509
6	₹1,67,19,789	₹2,17,12,000	₹49,92,211
7	₹1,84,23,088	₹2,46,64,000	₹62,40,912
8	₹2,01,26,386	₹2,76,16,000	₹74,89,614
9	₹2,18,29,684	₹3,05,68,000	₹87,38,316
10	₹2,35,32,982	₹3,35,20,000	₹99,87,018

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For more information, contact Absstem at absstem.com/contactus

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